PRAVEEN BUSHIPAKA

LLM Researcher • Machine Unlearning • Data-Centric AI • RAG & LLM Systems

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RESEARCH & ENGINEERING PROFILE

PhD researcher in Artificial Intelligence working at the intersection of LLM unlearning, data selection, trustworthy AI, retrieval, and production-oriented LLM systems. Research spans gradient-guided coreset selection, reliable unlearning evaluation, and privacy-preserving model adaptation; engineering work spans distributed training, PEFT, RAG, evaluation pipelines, and full-stack AI tooling. Interested in Research Scientist / Research Engineer / Applied Scientist roles where new NLP/LLM methods are translated into robust, measurable systems.

CORE TECHNICAL SKILLS

LLM Research & Alignment	Machine Unlearning, data selection / coresets, SFT, DPO, RLHF, GRPO, LoRA/QLoRA, PEFT, synthetic data, LLM-as-a-Judge
LLM Systems & Evaluation	PyTorch, Hugging Face Transformers, TRL, Accelerate, distributed/multi-GPU training, custom losses & datasets, benchmarking, error analysis
Retrieval & Applied NLP	Hybrid / Agentic RAG, embedding fine-tuning, reranking, GraphRAG, Milvus, ChromaDB, LangChain, LlamaIndex
Engineering & Deployment	Python, FastAPI, Docker, vLLM, Ollama, ONNX Runtime, TypeScript, SQL, experiment tracking, reproducible ML pipelines

RESEARCH EXPERIENCE

PhD in AI for Society | Scuola Superiore Sant'Anna & University of Pisa — **Nov 2023 – Present**

- Researching data-centric LLM unlearning: forget/retain set construction, gradient-guided selection, sampling strategies, privacy, and utility-preserving evaluation across models and unlearning algorithms.
- Developed GRACE, a gradient-guided coreset selection method for constructing compact forget and retain sets; accepted to Findings of EMNLP 2026.
- Designed an end-to-end Bank of Italy regulatory RAG system with embedding fine-tuning, hybrid retrieval, reranking, and automated evaluation using GPT-4o, Gemini Flash, and Llama-3.3-70B.
- Building ForgetLLM, a full-stack LLM unlearning and data-selection platform with modular algorithms, distributed training, experiment evaluation, and interactive model/metric workflows; research from the platform is being prepared for AAAI submission.

Researcher | Scuola Superiore Sant'Anna — **Sep 2022 – Aug 2023**

- Built legal-NLP pipelines for extractive summarization and rhetorical-role classification using Italian-Legal-BERT and Hugging Face; contributed to three peer-reviewed publications (ICAIL, ASAIL, CEUR).

Freelance ML Engineer | Rubidium Data, Germany — **Sep 2023 – Oct 2023**

- Built an automated ad-placement mapping pipeline for Ecosia using SERP data; fine-tuned BERT and benchmarked against GPT-3.5, reaching 95% accuracy.

Junior Data Scientist, Intern | Wunderman Thompson, Milan — **Dec 2019 – Feb 2021**

- Developed customer segmentation, recommendation, and analytics pipelines using clustering, market-basket analysis, Python, R, SQL, and Bash.

SELECTED PUBLICATIONS

GRACE: Gradient-guided Coreset Selection for LLM Unlearning
Praveen Bushipaka, Andrea D'Angelo, Lucia Passaro, Tommaso Cucinotta — Findings of EMNLP 2026 — Accepted

OARS — LLM Unlearning Research
Praveen Bushipaka et al. — Under review at EACL; data-centric LLM unlearning research

ForgetLLM: A Practical Implementation of LLM Unlearning
Praveen Bushipaka et al. — Under review at AAAI Demo

Standard vs. Modular Sampling: Best Practices for Reliable LLM Unlearning
Praveen Bushipaka, Lucia Passaro, Tommaso Cucinotta — ECML-PKDD Workshop / Springer CCIS, 2026

A Novel Multi-Step-Prompt Approach for LLM-based Q&As on Banking Supervisory Regulations
Daniele Licari, Canio Benedetto, Daniele Bovi, Praveen Bushipaka, et al. — Bank of Italy Occasional Paper 935, 2025

Legal Holding Extraction from Italian Case Documents using Italian-LEGAL-BERT Text Summarization
Daniele Licari, Praveen Bushipaka, et al. — ICAIL 2023

TALKS & RESEARCH COMMUNICATION

- ECML-PKDD 2026 — WIPE OUT Workshop: presented research on data selection and reliable machine unlearning for LLMs.

- Learning from Large, Complex and Structured Data Workshop, Scuola Superiore Sant'Anna: invited talk on machine unlearning methods for LLMs.

EDUCATION

PhD in AI for Society | Scuola Superiore Sant'Anna & University of Pisa — **2023 – Present**

MSc in Computer Science | University of Milan — **2017 – 2020**